

Rahul Satish

Ph.D. Researcher, ITU Copenhagen

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Education

Present Jan 2024	Department of Computer Science, IT University of Copenhagen Ph.D. in Computer Science (Expected to submit in March 2027) Advisor: Prof. Bernardo David Thesis: Towards Practical and Scalable Multiparty Protocols	Copenhagen, Denmark
Nov 2023 Sep 2021	Alexander Kofkin Faculty of Engineering, Bar Ilan University M.Sc. in Electrical Engineering (Computer Engineering Track) Advisor: Prof. Carmit Hazay Thesis: Circumventing Lower Bounds in Garbled Circuits	Ramat Gan, Israel
Aug 2021 Aug 2016	Birla Institute of Technology and Science (BITS) Pilani Dual Degree Programme, M.Sc.(Mathematics) & B.E.(Electronics & Instrumentation) Project: Exploring Round Optimal Solutions for the PSI-Sum with Cardinality Problem	Goa, India

Experience

Dec 2026 Sept 2026	Information systems Lab, Lucerne University of Applied Sciences and Arts  <i>Ph.D. Visiting Researcher</i> Advisor: Prof. Chen-Da Liu-Zhang Visiting the Information Systems Lab during my Ph.D.	Lucern, Switzerland
Feb 2026 Nov 2025	Social Informatics Laboratories, NTT Musashino R&D Center  <i>Ph.D. Intern</i> Advisor: Prof. Masayuki Abe Interning at NTT R&D, Social Informatics Laboratories	Tokyo, Japan
Oct 2021 Jul 2021	Ethereum Foundation Privacy and Scaling Explorations Group  <i>Consultant Researcher</i> Worked with a team developing a zk-rollup that directly supports the Ethereum Virtual Machine(EVM)	Remote
Dec 2021 Jul 2021	Chain Reaction Cryptography Research Group  <i>Consultant Researcher</i> Worked on modelling and understanding existing levelled and fully Homomorphic Encryption Schemes	Remote
Jun 2021 Jan 2021	QED-it Systems Cryptography Group  <i>Research Intern</i> Advisors: Daniel Benarroch , Michael Adjedj Worked on constructing efficient circuits for the BFV scheme, for comparing & sorting encrypted integers.	Remote
Aug 2020 May 2020	IBM Research Blockchain and Smart Contracts Group  <i>Research Intern</i> Advisors: Dr. Dhinakaran Vinayagamurthy , Nitin Singh Designed a modular framework to write interactive ZKP systems on arithmetic circuits	Bangalore, India

Publications*

S=In Submission, C=Conference, W=Workshop, P=Poster/Demo, J=Journal | [DBLP], [Scholar]

- [S.3] **Dual Adaptive Security for Garbling Schemes**
Anasuya Acharya, Carmit Hazay, [Rahul Satish](#)
[In Submission]
- [S.2] **Oblivious Transfer Extension for Dynamically-Scheduled Secure Computation**
Masayuki Abe, Peihao Li, Miyako Ohkubo, Octavio Perez-Kempner, [Rahul Satish](#), Mehdi Tibouchi
[In Submission]
- [S.1] **Eras MPC: Granular Preprocessing and Abort Recovery with Dynamic Committees**
Bernardo David, Ivan Damgard, [Rahul Satish](#)
[In Submission]
- [C.4] **Sublinear-Communication Layered MPC from HSS** 
Anasuya Acharya, Pierre Meyer, Divya Ravi, [Rahul Satish](#)
International Conference on the Theory and Application of Cryptology and Information Security [ASIACRYPT'26]

*indicates authors listed in alphabetical order following the Hardy-Littlewood rule.

- [C.3] **Oblivious Garbling and its Applications** [📄]
 Tomer Ashur, Carmit Hazay, [Rahul Satish](#)
International Conference on Security and Cryptography for Networks [SCN'26]
- [C.2] **The Landscape of Reusable Garbling** [📄]
 Anasuya Acharya, Carmit Hazay, [Rahul Satish](#)
International Conference on Security and Cryptography for Networks [SCN'26]
- [C.1] **Rumors MPC: GOD for Dynamic Committees, Low Communication via Constant-Round Chat** [📄], [📄], [🔗]
 Bernardo David, Arup Mondal, [Rahul Satish](#)
International Conference on the Theory and Application of Cryptology and Information Security [ASIACRYPT'25]
- [W.1] **Garbling 3-input AND gates** [📄], [🔗]
 Tomer Ashur, Carmit Hazay, [Rahul Satish](#)
Conference for Failed Approaches and Insightful Losses in Cryptology [CFAIL'23]

Talks

- “Sublinear-Communication from Layered MPC from HSS”
 > TPMPC 2026 [🌐] [📄] May 2026 (Aarhus, Denmark)
- “Rumors MPC: GOD for Dynamic Committees, Low Communication via Constant-Round Chat”
 > Asiacrypt 2025 [🌐] [📄] December 2025 (Melbourne, Australia)
- “Garbling gadgets and its applications to oblivious garbling”
 > TPMPC 2024 [🌐] [📄] June 2024 (Darmstadt, Germany)
- “Garbling 3-input AND gates”
 > CFail 2023 [🌐] [📄] August 2023 (Remote)

Select Technical Projects

- Generating custom polynomial constraints for efficient PLONK circuits** Jun'21 - Oct'21
 > Working on a zk-rollup that directly supports the Ethereum Virtual Machine(EVM). Prior to that, we were developing a ZKP layer over the EVM to be able to validate blocks through the PLONK proving system.
- Efficient sorting and comparison in the homomorphic setting** Jan'21 - Jun'21
 Advisors: [Daniel Benarroch](#), [Michael Adjedj](#) [🌐 Blog]
 > Identifying and implementing the best possible ways to construct HE circuits for comparison and sorting integer data.
- Non-Interactive Proof Generation from Interactive Zero Knowledge Protocols** May'20 - Aug'20
 Advisors: [Dr. Dhinakaran Vinayagamurthy](#), [Nitin Singh](#) [🌐 Repository]
 > Designed a modular framework for Interactive Zero Knowledge Protocols which was used to convert it to a non-interactive protocol. Implemented additional features for the design to support oracles, protocol composition, etc.. and tested existing protocols like Ligerio on it.

Academic Service

External Reviewer AC'26, Crypto'26, PKC'26, EC'26, PKC'25, AC'25, Crypto'25, PKC'25, ICDCS'25, TCC'24, EC'24, DCC'24, JoC'23, SAC'22, EC'22, CCS'21, TCC'21

Teaching and Leadership Roles

- Introduction to Cryptography, Cryptographic Computation and Blockchains** Courses Instructor at ITU
 > Responsibilities included evaluating tutorials, and helping students with the coursework and home assignments.
- IEEE ANTS, 2019, BITS Goa** Web Development Lead [Website] May'19 - Dec'19
 > Responsible for building and maintaining website of the IEEE International Conference on ANTS 2019.
- Introduction to Applied Cryptography, Quark Summer Technical Project** Instructor May'19 - Aug'19
 > Voluntarily mentoring students from across the nation to get used to cryptographic objects and the math behind it.

BITSkrieg, Ethical Hacking and PenTesting Club, BITS Goa *Co-ordinator*

Aug'18 - Jun'19

- Responsibilities include ensuring that the club works towards achieving its annually laid out goals.
- Handled a lecture series on Applied Cryptography for students of the college, being a member of the club.
- Involved in solving Capture the Flag(CTF) events such as Google CTF, PICO CTF, DEFCON regularly.

Quark Controls, BITS Goa *Core Member*

Mar'17 - Feb'18

- Successfully established contacts with companies like Mozilla, JP Morgan, LIGO for hosting workshops on campus.
- Responsible in helping the team arrange seminars and talks for the students on technical issues.

References

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- Prof. Bernardo David Associate Professor, IT University of Copenhagen, Denmark [🌐][✉]
 - Prof. Carmit Hazay Professor, Bar Ilan Univeristy, Israel [🌐][✉]
 - Prof. Masayuki Abe Senior Distinguished Researcher, NTT Social Informatics Laboratories, Japan [🌐][✉]
 - Dr. Tomer Ashur Co-Founder, 3MI Labs, Belgium [🌐][✉]
 - Daniel Benarroch CEO, Inversed Tech, Spain [🌐][✉]